NERS570 HW1

Kabelo Tsiane

September 2026

1 Introduction

Galaxy clusters are the largest gravitationally bound structures in the universe, and are the final output of large-scale structure formation. This makes them an excellent laboratory for studying astrophysical phenomena, such as galaxy formation, magnetohydrodynamics, and exotic dark matter interactions (Kravtsov and Borgani, 2012). On a cosmological scale they are used to track the abundance and distribution of baryonic matter to constrain cosmological models (Allen et al., 2011).

A key assumption of galaxy clusters in the ΛCDM cosmological model is the notion of self-similarity: the internal structure, dynamics and evolution only differs as a function of mass, and epoch (Kravtsov and Borgani, 2012). As such, measurements of the cluster mass are fundamental to any study with these systems. One of the most robust ways of doing this is through weak gravitational lensing: as light rays in the distant universe stream towards Earth, their paths are deflected by massive objects. These objects can be as small as planets and stars, all the way up to galaxies and galaxy clusters. By measuring the average shearing of distant galaxy images around the galaxy cluster, this allows us to measure the mass (Bartelmann and Schneider, 2001; Schneider et al., 2006).

2 Weak-Lensing Mass Measurement

In the weak-lensing regime, the azimuthally averaged tangential shear γ_t at projected radius R from the cluster centre is set by the excess surface mass density (Schneider et al., 2006, Part 2):

$$\gamma_t(R) = \frac{\Delta\Sigma(R)}{\Sigma_{\rm crit}}, \tag{1}$$

$$\Delta\Sigma(R) = \bar{\Sigma}(< R) - \Sigma(R). \tag{2}$$

The critical surface density depends only on geometry,

$$\Sigma_{\rm crit} = \frac{c^2}{4\pi G}\,\frac{D_s}{D_l\, D_{ls}}, \tag{3}$$

where D_l, D_s and D_{ls} are the angular diameter distances to the lens, to the source, and between them. Because source redshifts come from photometry, Eq. (1) is evaluated with the source-averaged inverse critical density for each tomographic bin b,

$$\left\langle \Sigma_{\rm crit}^{-1} \right\rangle_b = \frac{4\pi G}{c^2} \int_{z_l}^{\infty} {\rm d}z_s\; n_b(z_s)\, \frac{D_l\, D_{ls}}{D_s}, \tag{4}$$

with $n_b(z_s)$ the redshift distribution of bin b, normalised to unity. Fitting an NFW profile to the measured $\Delta\Sigma(R)$ then yields the cluster mass and concentration, for individual clusters or for stacks (e.g. Vitorelli et al., 2018). The accuracy of the result therefore rests on two properties of the source sample: an unbiased shear calibration, and well-characterised $n_b(z_s)$.

3 Making Source Selection in TXPipe Faster and More Robust

TXPipe (Prat et al., 2023) is the LSST DESC pipeline that takes shear and photometric catalogs to calibrated two-point measurements; its stages are run in parallel with MPI through `ceci`. For shear I use the

Algorithm 1 Chunked source selection with response accumulation

Require: catalogs $v \in \{\texttt{noshear}, \texttt{1p}, \texttt{1m}, \texttt{2p}, \texttt{2m}\}$; cuts; bin edges
assign each MPI rank a contiguous range of chunks (equal row counts)
for each chunk assigned to this rank **do**
    read the needed columns (e_1, e_2, S/N, size, photo-z) once, for all v
    for each catalog v **do**
        $m_v \leftarrow$ isfinite(all columns) $\wedge$ cuts; $b_v \leftarrow$ digitize(z_v, edges)
        write b_v (with -1 where $\neg m_v$) to the tomography catalog
        for each bin b **do**
            $N_{b,v}$ += $\#\{b_v = b\}$; $\mathbf{S}_{b,v}$ += $\sum_{b_v=b} \mathbf{e}$
        end for
    end for
end for
`Allreduce` all $N_{b,v}$ and $\mathbf{S}_{b,v}$ across ranks
on root: form $\langle e_i \rangle_{\pm j}$ and R_{ij} (Eq. 5); flag any bin with $N_{b,v} = 0$

metadetection scheme (Sheldon and Huff, 2017; Sheldon et al., 2020), in which objects are re-detected and re-measured on images sheared by $\pm\Delta\gamma$ along each axis, giving five catalogs: `noshear`, `1p`, `1m`, `2p`, `2m`. The shear estimate is

$$\hat{\gamma}_i = \left(\langle \mathbf{R} \rangle^{-1}\right)_{ij} \langle e_j \rangle, \qquad R_{ij} \simeq \frac{\langle e_i \rangle_{+j} - \langle e_i \rangle_{-j}}{2\,\Delta\gamma}, \tag{5}$$

where $\langle e_i \rangle_{\pm j}$ is the mean ellipticity over the objects that *pass the selection in the catalog sheared by* $\pm\Delta\gamma$ *in component* j. Selection effects are thus folded into R_{ij} automatically, but only if the source selection (cuts on signal-to-noise and size, and the tomographic bin assignment) is applied independently to all five catalogs, for every bin b. This is the work done by the `TXSourceMetadetect` stage, and it is the part of the pipeline I aim to improve.

Goals. *Speed*: for catalogs approaching 10^9 objects, the stage should be limited by I/O bandwidth rather than by repeated reads or serial post-processing. *Robustness*: it should give the same answer for any number of MPI ranks, and handle empty bins, non-finite values, and interrupted runs without silently corrupting Eq. (5). Algorithm 1 is the design I am implementing: each column chunk is read once for all five catalogs, only small integer counts and sums are reduced across ranks, and empty bins are flagged explicitly. I will verify it by profiling wall time and bytes read, measuring strong scaling with rank count, testing degenerate cases on toy catalogs, and checking that the output matches a reference run.

References

Steven W. Allen, August E. Evrard, and Adam B. Mantz. Cosmological Parameters from Observations of Galaxy Clusters. *Annual Review of Astronomy and Astrophysics*, 49(1):409–470, September 2011. ISSN 0066-4146, 1545-4282. doi: 10.1146/annurev-astro-081710-102514. URL `http://arxiv.org/abs/1103.4829`. arXiv:1103.4829 [astro-ph].

Matthias Bartelmann and Peter Schneider. Weak Gravitational Lensing. *Physics Reports*, 340(4-5):291–472, January 2001. ISSN 03701573. doi: 10.1016/S0370-1573(00)00082-X. URL `http://arxiv.org/abs/astro-ph/9912508`. arXiv:astro-ph/9912508.

Andrey Kravtsov and Stefano Borgani. Formation of Galaxy Clusters. *Annual Review of Astronomy and Astrophysics*, 50(1):353–409, September 2012. ISSN 0066-4146, 1545-4282. doi: 10.1146/annurev-astro-081811-125502. URL `http://arxiv.org/abs/1205.5556`. arXiv:1205.5556 [astro-ph].

J. Prat, J. Zuntz, et al. The Catalog-to-Cosmology Framework for Weak Lensing and Galaxy Clustering for LSST. arXiv:2212.09345, 2023.

Peter Schneider, Christopher S. Kochanek, and Joachim Wambsganss. *Gravitational Lensing: Strong, Weak and Micro*, volume 33 of *Saas-Fee Advanced Course*. Springer, Berlin, Heidelberg, 2006. Part 2 (weak lensing) by P. Schneider.

Erin S. Sheldon and Eric M. Huff. Practical Weak-lensing Shear Measurement with Metacalibration. *The Astrophysical Journal*, 841(1):24, 2017. arXiv:1702.02601.

Erin S. Sheldon, Matthew R. Becker, Niall MacCrann, and Michael Jarvis. Mitigating Shear-dependent Object Detection Biases with Metacalibration. *The Astrophysical Journal*, 902(2):138, 2020. doi: 10.3847/1538-4357/abb595. arXiv:1911.02505.

André Z. Vitorelli, Eduardo S. Cypriano, Martín Makler, Maria E. S. Pereira, Thomas Erben, and Bruno Moraes. On mass concentrations and magnitude gaps of galaxy systems in the CS82 survey. *Monthly Notices of the Royal Astronomical Society*, 474 (1):866–875, February 2018. ISSN 0035-8711. doi: 10.1093/mnras/stx2791. URL `https://doi.org/10.1093/mnras/stx2791`.